QUIZ #3 Sec 2

Consider the ODE:

$$x^2y'' + 4y' + x^2y = 0$$

Suppose φ is an analytic solution centered at $x=2$.

By computing some Taylor series & applying Ratio Test, find a lower bound for $\rho(\varphi)$, without solving φ.

(Given fact: $\frac{1}{x^2}$ is analytic around 2.)

Solution

$$y'' + \frac{4}{x^2} y' + 1\, y = 0 \quad (\square)$$

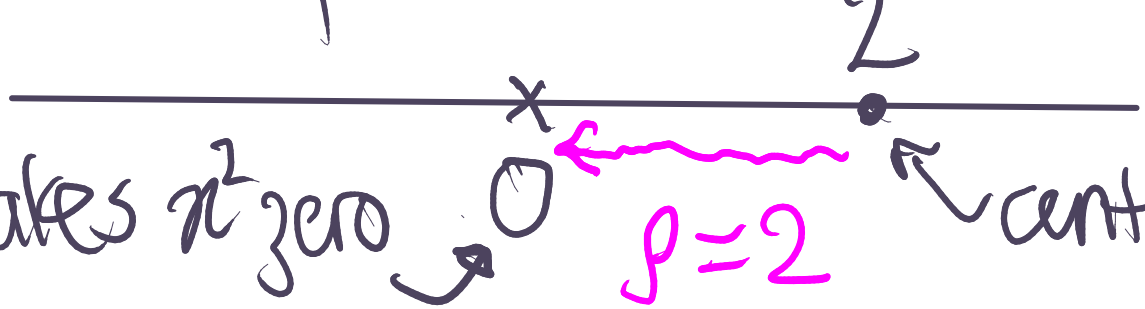
$\frac{4}{x^2}$ & 1 are analytic around $x=2$.

By theorem, the soln φ around 2 has $\rho(\varphi) \geq \min\left(\rho\left(\frac{4}{x^2}\right), \rho(1)\right)$

$$= \rho(4/x^2)$$

So we have to determine $\rho\left(\frac{1}{x^2}\right)$.

One way to do that is to look at the picture:



But we are not allowed to do that in this question. We're asked to determine the Taylor series of $f(x)=1/x^2$ at $x=2$ and then find $\rho(f)$ by Ratio Test. Observe:

$f^{(k)}(x)=(-1)^k\cdot(k+1)!\cdot x^{-2-k}$. So the Taylor series of f at $x=2$ is:

$$\sum_{k=0}^{\infty}\frac{f^{(k)}(2)}{k!}(x-2)^k=\sum_{k=0}^{\infty}\frac{(-1)^k(k+1)}{4\cdot 2^k}(x-2)^k$$

Now apply the Ratio Test: for absolute convergence

$$\lim_{k\to\infty}\left|\frac{A_{k+1}}{A_k}\right|=\lim_{k\to\infty}\left|\frac{k+2}{k+1}\,\frac{x-2}{2}\right|<1$$

$$\Leftrightarrow |x-2|<2 \Leftrightarrow \rho(f)=2$$

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You see: it's easy to compute the Taylor series for f by taking derivatives. There are other easy ways. For instance:

$$\frac{1}{x^2} = -\left(\frac{1}{x}\right)' = -\frac{d}{dx}\left(\frac{1}{2+x-2}\right) = -\frac{1}{2}\frac{d}{dx}$$

$$= -\frac{1}{2}\frac{d}{dx}\left(\frac{1}{1+\frac{x-2}{2}}\right)$$

$$= -\frac{1}{2}\frac{d}{dx}\sum_{k=0}^{\infty}(-1)^k\left(\frac{x-2}{2}\right)^k$$

$$= \sum_{k=1}^{\infty}\frac{(-1)^{k+1}}{2^{k+1}}\cdot k\,(x-2)^{k-1}$$

$$= \sum_{k=0}^{\infty}\frac{(-1)^k}{4\cdot 2^k}(k+1)(x-2)^k$$

exactly as found in the prev page!

Here & before, to be able to apply shuffling term-by-term derivation etc, we used the given fact that $\frac{1}{x^2}$ is analytic around $x=2$.